



Implications of Heightened Global Uncertainty for the East Asia and Pacific Region

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Macroeconomic, financial, and policy-related uncertainty have increased since the COVID-19 pandemic globally and in individual developing economies in the East Asia and Pacific region. Uncertainty shocks can transmit across borders and their economic consequence is quite sizeable, affecting both the financial sector and the real economy in the EAP region.

Introduction

Extensive literature documents heightened economic uncertainty since the COVID-19 pandemic (Cascaldi-Garcia et al. 2023). All types of uncertainty have increased: macroeconomic, financial, economic policy, geopolitical, and trade uncertainty. However, less is known about the implications of global uncertainty for emerging markets and developing economies (EMDEs) in the East Asia and Pacific (EAP) region. This Research and Policy Brief examines the effects of global uncertainty on EAP by addressing the following questions. First, what are the basic sources of economic uncertainty? Second, how has uncertainty evolved globally and in individual EAP EMDEs in recent decades? Third, how has global uncertainty affected macroeconomic and financial conditions in EAP?

The empirical results suggest that the effects of uncertainty—mainly related to the volatility (second moment) of the variables—are distinctive from the effects of the level (first moment) fluctuations in underlying business, financial, and policy variables. The economic consequences of the uncertainty shocks are quite sizeable, almost comparable to the effects of the “level” shocks. In addition, the results suggest that different types of uncertainty shocks affect EAP countries differently, individually and as a group cause somewhat heterogeneous effects on EAP countries. In terms of cumulative effect, macro uncertainty, followed by financial and economic policy uncertainty, contributes the most to the variations in the key domestic variables in EAP. In particular, macro uncertainty can explain up to one-quarter of variations in macro variables in EAP.

Evolution of global and EAP economic uncertainty

Global economic uncertainty rose sharply to a record-high level after the outbreak of the COVID-19 pandemic and stayed elevated through 2023 and early 2024. This Brief discusses the effect of global uncertainty on macroeconomic and financial conditions in the EAP. It highlights three types of uncertainty: *macroeconomic uncertainty*—forecast-based measures; *policy uncertainty*—news-based measures; and *financial uncertainty*—stock market volatility-based measures. Where a global index is unavailable, US uncertainty indexes are used as proxies for global uncertainty, as is common in the literature.

Box 1 presents more details about the measurement of different types of uncertainty.

Macroeconomic uncertainty is elevated in the United States, reflecting uncertain future output growth and inflation developments (figure 1, panel a). After peaking in April 2020 at the onset of the COVID-19 pandemic, it spiked again in 2022, mainly due to a surge in inflation. Despite the decline since mid-2022, inflation uncertainty is still high in the United States, reflecting disagreements on the speed and degree of the future (dis)inflation process. Global economic policy uncertainty, which aggregates future monetary, fiscal, and financial policies, is twice as high as the pre-pandemic levels (figure 1, panel b). In the United States, debates continue about the path of future monetary policy decisions by the Federal Reserve; the monetary policy uncertainty index has stayed persistently above the long-term averages since 2020. Similarly, financial uncertainty (VIX) surged in 2020 and 2022 and, despite declines, remains above pre-pandemic levels (figure 1, panel c).

Uncertainty regarding future output, inflation, financial markets, and economic policies is higher than before the pandemic in many EAP countries. Macro uncertainty, which suffered from record-high levels in 2020–21 (output uncertainty) or 2022 (inflation uncertainty), has generally declined since 2023 (figure 2). That said, uncertainty measures are at levels above pre-pandemic ones, and in some countries (Thailand), output and inflation uncertainty are still increasing. News-based economic policy uncertainty remains elevated—particularly in China. Survey-based uncertainty indicators on short- and long-term interest rates, which are typically a critical determinant of investment, have increased across all EAP economies (World Bank 2023).

Economic effects of heightened uncertainty

To assess the effects of global uncertainty on EAP, a series of structural vector autoregressive (SVAR) models are estimated over 2000–19 for five EAP countries: China, Indonesia, Malaysia, the Philippines, and Thailand. The models consist of a set of macro and financial variables in the United States (or global measures), China, and individual EAP countries. China, the largest EMDE, is taken separately in the model to reflect its role as both a source and a recipient of global uncertainty shocks. Three sources of uncertainty are distinguished:

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Acknowledgements: Nancy Morrison provided editorial assistance.

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Box 1. Empirical methodology and data

This Research and Policy Brief takes two sequential empirical steps: (1) measuring uncertainty; and (2) estimating SVAR models to quantify the economic effects of global uncertainty on EAP.

Measures of uncertainty

The Brief employs various measures of uncertainty based on the extensive literature and authors' calculations.

- **Forecast-based measures.** To gauge macroeconomic uncertainty, this set of measures relies on economic forecasting models for a variable of interest. For instance, a vector autoregressive (VAR) model is recursively estimated, and the standard deviation of the forecast error for the variable of interest is assumed to reflect the uncertainty in the variable.
- **Survey-based measures.** This set of measures uses economic surveys from professional forecasters on a variable of interest: output, inflation, short- and long-term interest rates, and current account. The uncertainty is measured as the survey responses' standard deviation or cross-sectional differences, such as inter-quartile ranges.
- **News-based measures.** To gauge policy uncertainty, this set of measures uses a keyword-searching algorithm. An example is the algorithm used by Baker et al. (2016) and Husted et al. (2020), which puts together economic policy uncertainty in the United States and other economies by searching the keywords for momentary, fiscal, and other policies in the local newspapers.

Structural vector autoregressive (SVAR) models

An open-economy model characterizes the data-generating process as follows:

$$AY_t = \sum_{i=1}^p B_i Y_{t-i} + \sum_{j=0}^q C_j X_{t-j} + \varepsilon_t.$$

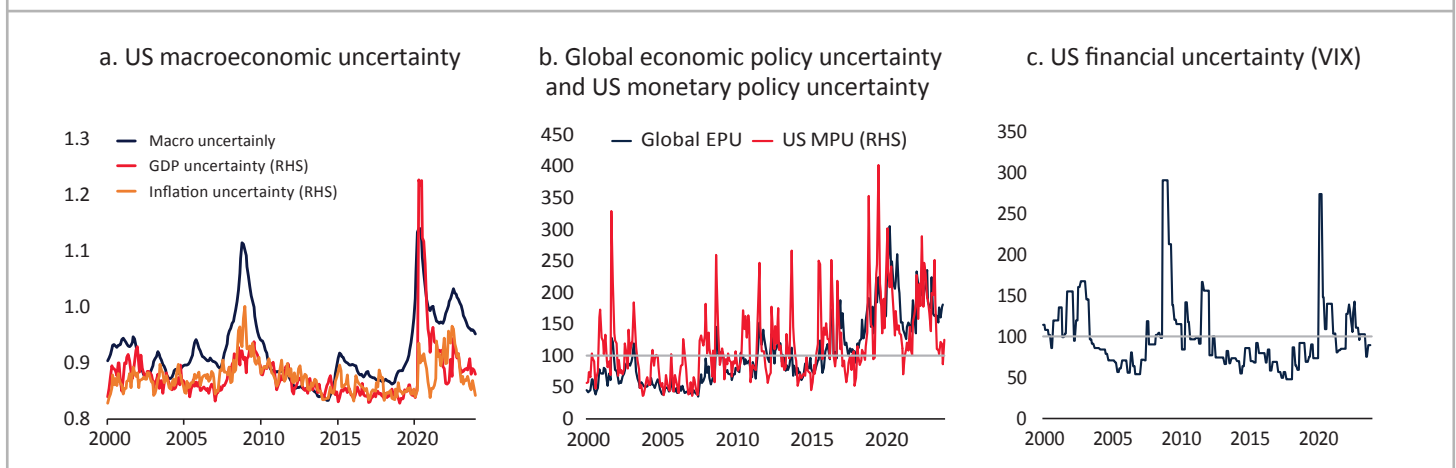
Y_t is an $n \times 1$ vector of endogenous macroeconomic and financial variables. X_t is exogenous regressors. A , B_i ($\forall i \geq 1$), and C_j ($\forall j \geq 0$) are nonsingular coefficient matrices. ε_t is an $n \times 1$ structural disturbances vector and serially uncorrelated.

Model. Both country-specific and panel SVAR models are run. Country models provide more detailed country-specific results, while the panel model is more efficient and clearer to present but loses country-specific results. This Brief reports the results based on panel SVARs. Country-specific results are featured as robustness or sensitivity checks. In identifying the uncertainty shocks, the Brief employs recursive restriction. Following the previous literature—such as Baker, Bloom, and Davis (2016); Leduc and Lui (2016)—and to overcome the endogeneity problem, the uncertainty measures are ordered *last* (along with various control variables to reduce the endogeneity).

Data set. The analyses are based on monthly macroeconomic and financial data for the global analysis, US analysis, and analysis of EAP countries in a panel setup. Where a global index is unavailable, US uncertainty indexes are used as proxies for global uncertainty, in keeping with the literature. The sample period is 2000–19. Main endogenous variables (in this order) are: US (global) output, inflation, uncertainty measure, domestic output, inflation, exchange rates, equity prices, and domestic uncertainty measure. Control variables are: volatility (VIX) index, US monetary policy instrument, time trends, and commodity prices. EAP countries in the sample include China, Indonesia, Malaysia, the Philippines, and Thailand. These countries were selected after considering the economic scale and the availability of uncertainty data. China, which is the largest emerging market and developing economy (EMDE), is taken separately in a model that includes both global (US) and Chinese variables. Three types of uncertainty measures are tested: macroeconomic, financial, and policy. For China's uncertainty measures, the dataset in Ahir et al. (2022) is also employed.

Robustness checks. The main empirical results are tested using alternative modeling [factor-augmented vector autoregressive (FAVAR) framework], alternative measures of uncertainty, different ordering of the variables, the inclusion of other control variables—in particular, the different types of uncertainty measures and the inclusion of post-pandemic data, among others. The results confirm that the key empirical results are robust to these alternative models.

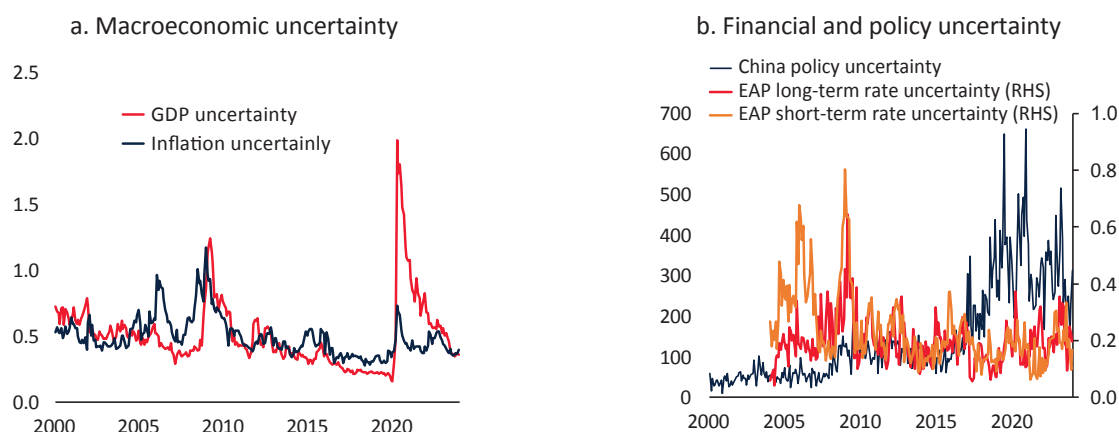
Figure 1. Evolution of uncertainty in the global economy and the United States



Source: World Bank estimates; Haver Analytics; Baker, Bloom, and Davis 2016; Jurado, Ludvigson, and Ng 2015.

Note: In panel a, macroeconomic uncertainty is based on forecast-based measures by Jurado, Ludvigson, and Ng (2015). The long-term average is 0.9. GDP and inflation uncertainty are based on a standard deviation of Consensus Forecasts for US GDP and Consumer Price Index (CPI) inflation. In panel b, the Global Economic Policy Uncertainty (EPU) and US Monetary Policy Uncertainty (MPU) index are based on news-based measures by Baker, Bloom, and Davis (2016). In panels b and c, the horizontal lines indicate the long-term average (set to 100). RHS = right-hand side.

Figure 2. Evolution of economic uncertainty in the East Asia and Pacific region



Source: World Bank estimates; Consensus Economics; Baker, Bloom, and Davis 2016.

Note: In panel a, EAP macroeconomic uncertainty is based on the average cross-sectional standard deviation of Consensus Forecast surveys for four emerging markets and developing economies (EMDEs) in EAP (China, Indonesia, Malaysia, Thailand). In panel b, China policy uncertainty is based on news-based measures by Baker et al. (2016). EAP financial uncertainty is based on the average standard deviation of Consensus Forecasts for 3-month Treasury bill yields and 10-year bond yields in the four EAP countries. EAP = East Asia and Pacific; RHS = right-hand side.

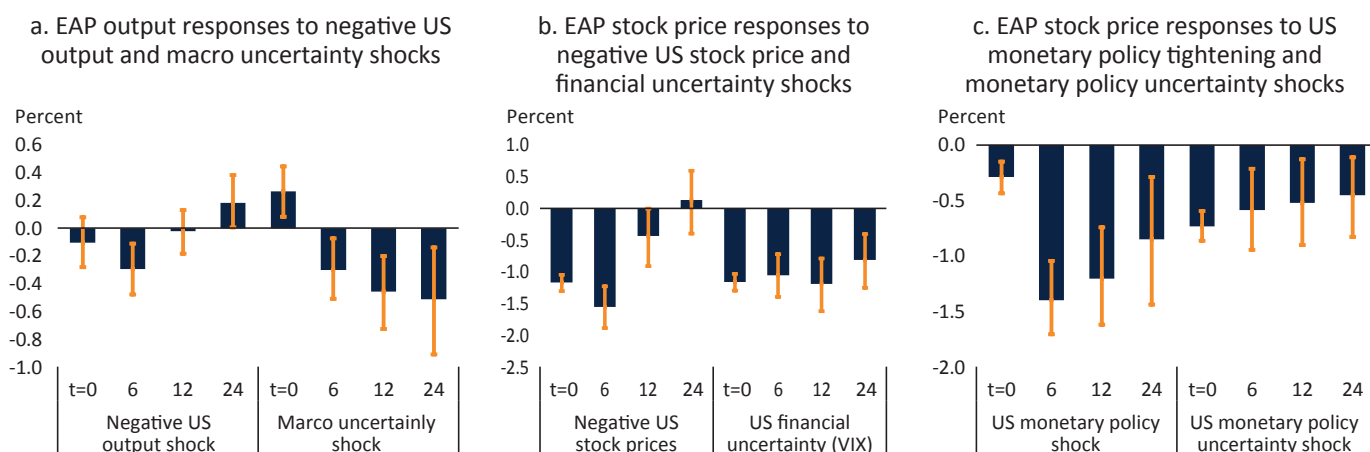
macroeconomic, financial, and policy (for estimation details, see box 1).

The findings suggest that global uncertainty shocks have negatively affected macro and financial conditions in EAP countries. This result is statistically and economically significant based on average results (that is, based on the panel model), as well as in country-specific perspectives—the results hold for all EAP countries in the sample. Following a one-standard-deviation increase in US macro uncertainty, growth of output declined by 0.5 percent and stock prices declined by 3.0 percent in EAP within one year after the shock (figure 3, panels a and b). The effects are stronger than the effects from a one-standard-deviation decline in US output (by 0.3 percent and 1.0 percent, respectively). Global financial market and economic policy uncertainty also had significant

impacts on key domestic financial variables in EAP, although their impacts are overall more immediate but short-lived compared with macro uncertainty (figure 3, panel c). Following a one-standard-deviation VIX shock, average stock prices in EAP declined by 1.2 percent, which is comparable to the effects of a negative shock on US equity prices (-1.6 percent). The effects of policy uncertainty shocks appear to add another layer of negative effects on EAP countries. While US monetary tightening led to a decline in EAP stock prices by up to 1.4 percent, heightened US monetary policy uncertainty resulted in another 0.7 percent decline in stock prices.

The sizeable uncertainty effects reflect various channels of international transmission of uncertainty shocks. First of all, the dynamic responses of the macro variables are consistent with the theories on the economic effects of uncertainty—typically

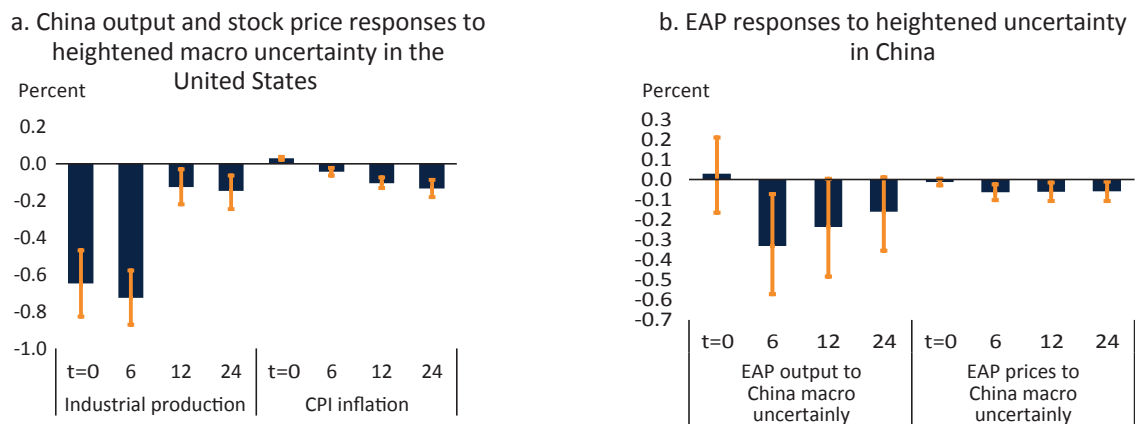
Figure 3. Impact of US uncertainty on the East Asia and Pacific region



Source: World Bank estimates; Jarociński and Karadi 2020.

Note: The results are based on panel vector autoregressions estimated with a sample for January 2000 through December 2019 for the United States and four emerging markets and developing economies (EMDEs) in EAP (Indonesia, Malaysia, Philippines, Thailand). The model includes, in this order, the following measures: the US production, US Consumer Price Index (CPI), US uncertainty; and domestic (EAP) industrial production, prices, stock prices, exchange rates, and uncertainty. Panel a shows dynamic responses of EAP industrial production to a one-standard-deviation increase in macroeconomic uncertainty and decline in output. Bars indicate the median responses, and vertical lines indicate 16 percent–84 percent confidence intervals. Panel b shows dynamic responses of EAP equity prices to a one-standard-deviation increase in financial uncertainty (VIX) and decline in US equity prices. Panel c shows dynamic responses of EAP stock prices to US monetary policy tightening shock and US monetary policy uncertainty shock. EAP = East Asia and Pacific.

Figure 4. Impact of United States and China uncertainty on the East Asia and Pacific region



Source: World Bank estimates.

Note: The results are based on a panel Vector autoregression estimated for January 2000 through December 2019 for four emerging markets and developing economies (EMDEs) in EAP (Indonesia, Malaysia, Philippines, Thailand). The model includes, in this order, the US production, US Consumer Price Index (CPI), US uncertainty measure; China production, China CPI, China uncertainty measure; and domestic (EAP) industrial production, prices, stock prices, exchange rates, and uncertainty measures. In panel a, bars show dynamic responses of China industrial production and CPI to a one-standard-deviation increase in macroeconomic uncertainty in the United States. In panel b, bars show dynamic responses of EAP production to a one-standard-deviation increase in macroeconomic uncertainty in the United States and China. EAP = East Asia and Pacific.

on the “wait-and-see” channel that strains private investment and consumption (Binder 2017; Bloom 2009). In particular, high levels of macro uncertainty have a negative impact on investment growth, as investors become hesitant to invest in new projects or expand their existing businesses (Leduc and Liu 2016). Second, the reactions of financial asset prices and increases in credit costs also contribute to the contractionary effects of economic uncertainty. Third, following the heightened global and US uncertainty, the uncertainty indicators in individual EAP countries also responded significantly, which supports the narratives of internationally correlated risk and uncertainty (Londono et al. 2021). The heightened uncertainty in the domestic economy is estimated to have played a negative role in economic confidence/sentiments in EAP countries (Ha and So 2023).

Estimation results also suggest that while China was negatively affected by the global and US uncertainty shocks, its own uncertainty shocks had an impact on the macroeconomic and financial conditions in EAP (figure 4). Following a one-standard-deviation increase in macro uncertainty in China, industrial production in EAP countries declined by around 0.3 percent—about half of the responses to the same type of US uncertainty. Again, China's heightened policy uncertainty was followed by heightened uncertainty in EAP countries. These results collectively suggest that China is both a source and

recipient of uncertainty shocks in the global markets, particularly in connection with EAP countries.

Conclusion

The findings lead to at least two types of novel implications. First, the effects of uncertainty—which is mainly related to the volatility (second moment) of the variables—are distinctive from the effects of the level (first moment) fluctuations in underlying business, financial, and policy variables. The economic consequence of the uncertainty shocks is quite sizeable, almost comparable to the effects of the “level” shocks. These findings are quite robust across different types of uncertainty measures—specifically, across macro, financial, and policy uncertainties. Second, the results suggest that different types of uncertainty shocks affect EAP countries differently, individually and as a group. In terms of cumulative effect, macro uncertainty, followed by financial and economic policy uncertainty, contributes the most to the variations in the key domestic variables in EAP. In particular, macro uncertainty, which explains up to 40 percent of total variations in US output and prices, can explain up to one-quarter of variations in macro variables in EAP. This result is in line with the findings of Stock and Watson (2012), who argue that the global uncertainty shock has been a key driver of global recessions and other severe fluctuations.

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